

ANNUAL PERFORMANCE AND ACHIEVEMENTS REPORT (JANUARY 2025 – PRESENT)

R. SELVAKUMAR, M.E., (Ph.D)

Assistant Professor-AI & DS ➡ <https://selvakumar83.github.io/Portfolio/>

1. Academic and Research Profile

- Area of Specialization: Machine Learning and Cryptography
- Ph.D Status: Course Work Completed and Qualified

2. Professional Experience

- 2 Years – Industry Experience
- 14 Years 10 Months – Teaching Experience
- Currently serving at St. Martin's Engineering College
(Since 06 January 2025 –Present, 1 Year 3 Months)

3. Teaching Excellence (2025 – Present)

- Subjects Handled:
 - Machine Learning
 - Cryptography and Network Security
- Academic Performance:
 - Machine Learning – 94.3% Pass Percentage
 - Cryptography and Network Security – 97.6% Pass Percentage

4. Student Development and Mentorship

- Supervised 7 Undergraduate Projects in Machine Learning
- Serving as Class In-Charge and Mentor for academic and career guidance
- Supporting students in technical, aptitude, and interview preparation

5. Innovative Teaching and Digital Initiatives

- Developed an AI-Based Classroom Application (ChatGPT-inspired basic model) using Streamlit and GitHub ➡ <https://selva-chatbot-aids-selvakumarownership.streamlit.app/>
- Designed and deployed a Machine Learning Quiz Application using Streamlit Cloud
- Promoting self-learning through interactive and digital platforms

6. Technical Proficiency

- Tools and Platforms: Anaconda, Streamlit, GitHub
- Strong focus on practical and application-oriented teaching methodologies

7. Research and Intellectual Contributions

- Published 1 Research Article in a Scopus-Indexed Journal
- Published 2 Patents in emerging technology domains

8. Placement and Coordination Responsibilities

- Serving as Department Placement Coordinator (Last 7 Months)
- Coordinating with Training & Placement Cell and Industry Recruiters
- Organizing and facilitating campus recruitment drives
- Acting as a liaison between students, department, and placement cell

9. Student Engagement and Academic Support

- Regularly updating students with latest technologies and industry trends
- Encouraging active participation in academic and placement activities
- Collecting and analyzing student feedback to enhance teaching effectiveness

10. Key Strengths and Contributions

- Technology-driven teaching approach
- Strong commitment to student success and employability
- Effective academic coordination and leadership
- Continuous contribution to research and innovation

SELVAKUMAR R

Assistant Professor

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WWW: <https://selvakumar83.github.io/Portfolio/> |

PROFESSIONAL SUMMARY

Dynamic Assistant Professor with over 14 years of experience teaching computer science engineering principles across diverse educational levels, complemented by 2 years of industry experience. Recognized for a strong commitment to fostering student engagement and motivation, with expertise in both computer hardware and software. Proficient in developing comprehensive course syllabi and integrating innovative technologies into the curriculum to enhance learning outcomes. Dedicated to continuous professional growth, ensuring a high-quality educational experience that empowers all students to succeed.

SKILLS

- Machine Learning
- Streamlit, GitHub, Anaconda
- Curriculum Planning & Teaching Methodologies
- Student assessment
- Mobile app development
- Student mentoring
- Relationship building
- Cryptography & Network Security
- AI Application Development
- Programming languages
- Video editing
- Website publication
- Data analysis

WORK HISTORY

Assistant Professor

St.Martin's Engineering College

Jan 2025 - Current

Secunderabad, India

- Teaching Machine Learning and Cryptography & Network Security
- Achieved 94.3% and 97.6% pass percentages respectively
- Introduced new teaching methodologies, such as flipped classrooms and project-based learning, enhancing student comprehension of complex topics.
- Evaluated student performance through exams and assignments, providing constructive feedback to foster continuous improvement.
- Advised students on their capstone projects, ensuring successful completion and showcasing their skills to potential employers.
- Demonstrated execution of Machine Learning programs from GitHub on mobile devices using cloud-based development environments such as GitHub Codespaces integrated with Visual Studio Code, enabling flexible and accessible hands-on learning.
- RSA Algorithm in Cryptography – Lecture recording available on YouTube:
<https://www.youtube.com/watch?v=oWxDf-v0Yf8&t=1335s>

Assistant Professor

Aug 2011 - Dec 2024

Dhanalakshmi Srinivasan Engineering College

Perambalur, India

- Enhanced student understanding of computer science concepts by incorporating real-world examples and case studies into lectures.
- Advised students on their capstone projects, ensuring successful completion and showcasing their skills to potential employers.
- Organized workshops for students to build essential skills in programming languages, software tools, and emerging technologies.
- Served on department committees to improve curriculum design, faculty recruitment, and program accreditation processes.
- Established productive relationships with fellow faculty members by sharing resources, collaborating on joint projects, and providing peer support.
- Evaluated student performance through exams and assignments, providing constructive feedback to foster continuous improvement.

Programmers

Feb 2006 - Mar 2008

IGATE-IT Company

Chennai, India

- Tested and deployed scalable and highly available software products.
- Modified existing software to correct errors, adapt to new hardware and improve performance.
- Evaluated multiple software solutions during early software architecture plotting and system migration planning stage.
- Collaborated with product management to design, build, and test systems.
- Rapidly prototyped new data processing capabilities to confirm integration feasibility into existing systems.
- Developed next generation integration platform for internal applications.

EDUCATION

Doctor of Philosophy (Ph.D): Enhancing the security in cryptosystem by generating the key based on Artificial Neural Networks

Expected in Jan 2027

Bharathidasan University

Tiruchirappalli

Completed Coursework:

- [RESEARCH AND PUBLICATION ETHICS] [June-2024]
- [MATHEMATICAL CRYPTOGRAPHY] [June-2024]
- [MACHINE LEARNING] [June-2024]

Master of Engineering: Computer Science Engineering

Jun 2011

Anna University of Technology

Coimbatore

Density Conscious Subspace Clustering for High-Dimensional Data (Capstone Project)

Density-conscious subspace clustering for high-dimensional data involves discovering clusters from different dense thresholds in various subspace cardinalities using the DENcos algorithm and genetic algorithm.

Master of Science: Computer Science

Jun 2005

Jamal Mohamed College

Tiruchirappalli

Machine Learning Approach for Database Integrity Verification and Rule Discovery (Capstone Project)

- Developed a machine learning-based system to verify database integrity and detect inconsistencies in large datasets.
- Applied classification, association rule mining, and sequence analysis for pattern discovery.
- Automated data validation process, reducing manual effort and improving accuracy.
- Utilized tools such as Python and machine learning libraries for implementation.

Higher Secondary Certificate (HSC) +2

May 2000

Nehru Higher Secondary School

Eraiur, Perambalur

Secondary School Leaving Certificate (SSLC) 10th

May 1998

Nehru Higher Secondary School

Eraiur, Perambalur

PUBLICATIONS & PATENTS

- Odd-Determinant Matrix Pattern Recognition for Secure Hill Cipher Key Generation Using Artificial Neural Networks, *STAPS Journal*.
This paper explores the integration of Artificial Neural Networks (ANNs) to enhance Hill Cipher key generation using odd-determinant matrix pattern recognition, improving cryptographic security and computational efficiency.
- Content-Based Image Retrieval Using Continuous Feature Selection, *International Journal of Scientific Research and Modern Education (IJSRME)*, ISSN: 2455-5630, Volume I, Issue I. Focuses on efficient feature extraction techniques to improve content-based image retrieval systems.
- Privacy-Preserving Data Sharing with Anonymous ID Assignment, *International Journal of Current Research and Modern Education (IJCRME)*, ISSN: 2455-5428, Volume I, Issue I. Proposes a secure data-sharing framework ensuring user privacy through anonymous ID assignment.
- Elevating File Security Through Multiple Image Steganography, Indian Patent Office, Patent No. 202541039000 A. Developed a method to secure confidential files by embedding data across multiple images using steganography, with optional encryption for enhanced protection.
- Intelligent Flood Monitoring and Alert System Using CNN, Indian Patent Office, Patent No. 202541039008 A. Designed a system using Convolutional Neural Networks (CNN) for real-time flood prediction and alert generation, supporting effective disaster management.

ACADEMIC ACTIVITIES

- **Association Coordinator:**
Experienced in organizing and managing academic associations, including effective event planning and coordination.
- **Department Maintenance In-charge:**
Oversaw departmental facilities and resources, ensuring efficient maintenance, safety compliance, and optimal functionality.
- **Head of the Department:**
Proven track record in leading academic departments, including curriculum development and faculty coordination, with strong strategic planning and decision-making skills to drive growth and excellence in teaching and research.
- **Class Advisor:**
Mentored students by tracking academic progress and guiding them to appropriate resources, while also overseeing events, resolving conflicts, and facilitating effective communication.

- **Time Table In-charge:**

Coordinated course schedules, optimized faculty and classroom allocation, and ensured smooth academic planning and execution.

- **Achievement:**

Received a Certificate of Appreciation from IQAC, St. Joseph's College (Autonomous), Tiruchirappalli (2022) for designing and developing a value-added course on *Scratch Programming* through JosTEL LMS, including the creation of 9 e-content modules with a total duration of 3 hours 1 minute.

FACULTY DEVELOPMENT SEMINARS

- Faculty development program on Data Warehousing and Business Intelligence held on January 28, 2010, at Bharathiar University, Coimbatore
- The Middleware Technologies workshop was held on July 8-9, 2011, at Sri Ramakrishna Engineering College in Coimbatore.
- Faculty Development Program on 'Introduction to Mobile Computing' held on December 16th and 17th, 2015 at Dhanalakshmi Srinivasan Engineering College, Perambalur.
- AICTE-QIP sponsored a two-week short-term course on 'Deep Learning with Applications in Language, Vision, and Control' held from December 4th to 17th, 2017, at Thiagarajar College of Engineering in Madurai.
- Participated in a Faculty Development Program on Modern Web Development & AI Integration under the *Next Gen Employability Program*, conducted by Edunet Foundation in collaboration with TASK (Telangana Academy for Skill and Knowledge), from 01 December 2025 to 12 December 2025.
- Attended a 5-day Faculty Development Program (FDP) on *AWS Cloud Practitioner*, organized by Holy Cross College (Autonomous), Tiruchirappalli in collaboration with ICT Academy of Tamil Nadu (19–23 August 2024).

CERTIFICATION

- Jul 2008 Cisco Networking Academy-Fundamentals of Java Programming

ACCOMPLISHMENTS

- Organized a Webinar on Web Application Development using CodeIgniter Framework, DSEC, Perambalur, June 2021.
- Organized a Webinar on Software Testing, DSEC, Perambalur, June 2021.
- Organized a Webinar on Android Application Development, DSEC, Perambalur, June 2021.
- Organized an Intra-Departmental Symposium, DSEC, Perambalur, February 2020.
- Organized a Workshop on Recent Trends in IT, DSEC, Perambalur, September 2019.
- Organized a National-Level Technical Symposium, DSEC, Perambalur, September 2018.
- Organized a State-Level Technical Symposium, DSEC, Perambalur, September 2014.
- Organized a Two-Day Faculty Development Programme on Open Source Software, DSEC, Perambalur, December 2011.
- Organized a Webinar on Data Science and Machine Learning Applications, DSEC, Perambalur, August 2022.
- Organized a Workshop on Artificial Intelligence and Deep Learning, DSEC, Perambalur, September 2023.
- Organized a National-Level Seminar on Emerging Trends in AI & Data Analytics, DSEC, Perambalur, January 2024.

PROFESSIONAL AFFILIATIONS

- Indian Society of System for Science & Engineering (ISSE) Life Member-LM06034

INTERESTS

- Research in AI & Machine Learning
- Educational Technology

PROJECTS

- Developed an AI-based classroom application (ChatGPT-inspired) using Streamlit and GitHub
- Designed and deployed a Machine Learning Quiz Application using Streamlit Cloud

COMMUNITY OUTREACH AND INVOLVEMENT

- Tamil Nadu State Level Youth Convention held from December 4th to 6th, 2009, at Ramakrishna Mission Vidyalaya in Coimbatore.,
- DSEC-Cleaning Campaign on the eve of Dussehra, held on September 27, 2017, at Dhanalakshmi Srinivasan Engineering College, Perambalur.

WEBSITES / PORTFOLIO / PROFILES

- <https://selvakumar83.github.io/Portfolio/>

DECLARATION

I hereby declare that the information provided above is true to the best of my knowledge, and I accept responsibility for the accuracy of the details given.



Mr. R. SELVAKUMAR



Reg No. 091111100033



The Syndicate of the Anna University Coimbatore hereby makes known that
SELVAKUMAR R *has been admitted to the* **DEGREE OF MASTER**
OF ENGINEERING *in* **COMPUTER SCIENCE AND**
ENGINEERING *under the Faculty of Information and Communication*
Engineering having completed the prescribed programme of study through the
University Department *and having been certified by duly appointed*
examiners to be qualified to receive the same and having been placed by them in
the FIRST CLASS *at the Examination held in* **June 2011.**

ANNA UNIVERSITY COIMBATORE ANNA UNIVERSITY COIMBATORE ANNA UNIVERSITY COIMBATORE ANNA



Coimbatore 641 047
India
March 2012

Controller of Examinations

Registrar

Vice-Chancellor

Given under the Seal of the University